

Qatar University
College of Arts and Sciences
Department of Mathematics and Statistics

STAT 481 — Practice Sheet 1

Data, Summary, Display, Distance

Sun, Sep 6, 2026 • in class, 50 minutes • Lectures 1–5

Instructor: Prof. Dhafer Malouche • not graded

How to read a solution — read this first

A printed solution is a record of somebody else's thinking; reading it is not producing it. Work each part *first*; stuck for ninety seconds, read only the solution's *first line*, then carry on and ask: “which line of mine would have broken if the numbers had been different?” The [tag] on each part is your revision index: the lecture, and the kind of task.

The data. worldhealth2022, the file of Lectures 2–5, loaded exactly as the lectures load it. **You have seen it before** — deliberately: this sheet revises the code you ran, on the numbers you ran it on. Thursday's quiz will not.

Problem 1 — From the file to the two matrices [L1 → L2 → L3] *load, inspect, order, $\bar{x}$, S, R*

(a) [L1 · R-OUT] The session opens the file the way Lecture 3 did:

```
[1] 175 11
'data.frame': 175 obs. of 11 variables:
 $ country      : chr  "Afghanistan" ...
 $ region       : chr  "Middle East, North Africa, Afghanistan & Pakistan" ...
 $ income       : chr  "Low income" ...
 $ life_exp     : num  65.6 ...
 $ infant_mort  : num  51.6 8.3 ...
 $ fertility    : num  4.93 ...
 $ pop65       : num  2.35 ...
 $ urban_pct    : num  25.4 ...
 $ gdp_pc       : num  357 ...
 $ health_exp_pct: num  23.1 ...
 $ internet_pct : num  15.9 ...
```

State n and p . Say what one row of $\mathbf{X}$ stands for. Name the columns that are *not* variables, and say why the lectures always write `wh[, 4:11]` rather than `wh`.

(b) [L1 · R-PREDICT] We put the country names on the rows, as Lecture 5 does, and take one indicator for the six Gulf countries:

```
W <- wh[, 4:11] # the eight indicators; columns 1-3 are labels
rownames(W) <- wh$country # country NAMES on the rows, as in Lecture 5
gcc <- c("Bahrain", "Kuwait", "Oman", "Qatar", "Saudi Arabia",
        "United Arab Emirates")
v <- W[gcc, "infant_mort"]; names(v) <- gcc
v
```

Bahrain	Kuwait	Oman
7.0	7.4	8.5
Qatar	Saudi Arabia	United Arab Emirates
5.0	5.0	4.2

Before running anything, write down what each of `sort(v)`, `order(v)`, `rank(v)` and `rank(-v)` returns — values, positions or standings; which end carries rank 1 in each of the two `rank` calls; and how the two countries that share a value will print. Then check against the block below.

United Arab Emirates	Qatar	Saudi Arabia
4.2	5.0	5.0
Bahrain	Kuwait	Oman
7.0	7.4	8.5
[1] 6 4 5 1 2 3		
Bahrain	Kuwait	Oman
4.0	5.0	6.0
Qatar	Saudi Arabia	United Arab Emirates
2.5	2.5	1.0
Bahrain	Kuwait	Oman
3.0	2.0	1.0
Qatar	Saudi Arabia	United Arab Emirates
4.5	4.5	6.0

(c) **[L2 · by hand]** Three countries on `fertility` and `urban_pct`, rounded to whole numbers so that the arithmetic is exact — `round(W[rows, cols], 0)` with the three names and the two indicators prints

	fertility	urban_pct
Oman	2	80
United Arab Emirates	1	86
Jordan	3	92

Take this 3×2 table as $\mathbf{X}$. Compute $\bar{\mathbf{x}}$, $\mathbf{S}$, r and $|\mathbf{S}|$ exactly, keeping every quantity as a fraction. Say in one sentence what the divisor $n - 1$ is doing.

(d) **[L2 · R-OUT]** As Lecture 2 printed them: the diagonal of $\mathbf{S}$, then $\mathbf{R}$ at `options(width = 100)`, all 175 countries.

life_exp	67.3	infant_mort	517.7	fertility	1.6	pop65	51.0	urban_pct	487.4	gdp_pc	801635374.1
health_exp_pct	10.7	internet_pct	648.6								
life_exp	1.00	infant_mort	-0.92	fertility	0.66	pop65	0.55	urban_pct	0.56	health_exp_pct	0.17
infant_mort	-0.92	1.00	0.78	-0.56	-0.46	-0.39	-0.15	-0.74			
fertility	-0.81	0.78	1.00	-0.67	-0.50	-0.42	-0.21	-0.84			
pop65	0.66	-0.56	-0.67	1.00	0.43	0.59	0.36	0.61			
urban_pct	0.55	-0.46	-0.50	0.43	1.00	0.52	0.23	0.72			
gdp_pc	0.56	-0.39	-0.42	0.59	0.52	1.00	0.16	0.52			
health_exp_pct	0.17	-0.15	-0.21	0.36	0.23	0.16	1.00	0.20			
internet_pct	0.77	-0.74	-0.84	0.61	0.72	0.52	0.20	1.00			

- Off the diagonal, name the strongest association in $\mathbf{R}$ and the weakest, with values, and say in words what the sign of the strongest means.
- The eight variances add up to $\text{tr}(\mathbf{S})$. Why does `gdp_pc` own 100% of it to two decimals, and what happens to that share if income is recorded in *thousands* of dollars (`gdp_pc / 1000`)? Direction and reason; a number is a bonus.

(e) **[L3 · R-WRITE]** Write the R that prints the mean of *each* of the eight indicators within *each* income group, one table, groups in economic order (Lecture 3's `econ` vector). Then name the one indicator whose four group means do not run in one direction as income rises.

(f) **[L2 · concept]** An economist wants to know how many dollars of income go with one extra year of life expectancy; a journalist wants to know which two of the eight indicators are most tightly linked. Which matrix does each one get, and why? Then: two pairs have $r = 0.6$ and $r = -0.6$ — what does $|\mathbf{R}| = 1 - r^2$ say about them, and what does it miss?

Problem 2 — Which countries are unusual, and by which yardstick? **[L4 → L5 · R only]**

write the R, run it, read the answer off the console

Every part of this problem is answered by writing R and running it. Each question says what to compute; write the code, run it, and answer in one or two sentences that quote the numbers the console gave you. Laptops open.

(a) **[L3 → L5 · R-WRITE]** Build the matrix `X` of the eight indicators with the country *names* on its rows, after dropping the one country whose life-expectancy cell Lecture 3 (Step 5) taught you to drop. Report the dimensions of `X` and the standard deviation of each column to two decimals. Which column would drive a straight-line distance computed on the raw columns, and roughly how many times larger is its spread than the smallest?

(b) **[L4 · R-WRITE]** Compute the correlation between `life_exp` and `gdp_pc`, then between `life_exp` and `log(gdp_pc)`. Which is larger? Say what the difference tells you about the *shape* of the `life_exp`–`gdp_pc` cloud, and name the Lecture 4 display that would show that shape directly.

(c) **[L5 · R-WRITE]** For the six Gulf countries of Problem 1(b) (Bahrain, Kuwait, Oman, Qatar, Saudi Arabia, United Arab Emirates), compute the straight-line distance matrix on the raw columns of `X`. Then standardise *all* 174 rows with `scale()`, take the same six rows and compute it again. For each matrix, name the closest pair and the farthest pair. Which answer changed, and why?

(d) **[L5 · R-WRITE]** For all 174 countries compute the Mahalanobis distance d^2 from the centre $\bar{\mathbf{x}}$ of the eight-indicator cloud, and the raw straight-line distance from the same centre. Print the six farthest countries on each, and Qatar's standing on each. Then print the names of the six richest countries by `gdp_pc`. Which of your two lists is that, and what does it tell you about which list to trust?

(e) **[L5 · R-WRITE]** Using the standardised table `Z` of part (c), compute the straight-line distance of every country from the centre of `Z` and print the six farthest; compare them with the d^2 list of part (d). Then use `all.equal()` to check whether the Mahalanobis distances computed on `Z` equal those computed on `X`. What do the two results say about the sentence “standardising the columns turns the Euclidean distance into the Mahalanobis distance”?

Problem 3 — Fill in the blanks [L2 → L5 · quiz rehearsal] laptops closed

If the clock reads 33 and this problem has not started, skip to it. The transcript below is one R session on `worldhealth2022`, run exactly as Lectures 2–5 ran it; every line is correct as printed, and pieces of it are hidden behind numbered yellow boxes. Write what each box hides — a function name, an argument, a number or a name — in the answer table on the next page. Everything you need is on this page: a hidden number is always one that another line of the transcript, with what you know from Lectures 1–5, lets you recover. One point per box.

```
> wh <- read.csv("data/worldhealth2022.csv", (1))
> dim(wh)
[1] 175 11
> ok <- wh$life_exp > 40 # Lecture 3, step 5: drop the broken row
> sum(ok)
[1] (2)
> X <- (3)(wh[ok, 4:11])
> rownames(X) <- wh$country[ok]
> dim(X)
[1] 174 (4)
> round((5)(X), 2)
      life_exp infant_mort fertility pop65 urban_pct
      72.83    18.91      2.44    9.78    61.73
      gdp_pc health_exp_pct internet_pct
      18554.28      6.80      69.89
> round(cov(X)[1:4, 1:4], 2)
      life_exp infant_mort fertility pop65
life_exp    50.90    -113.70    -7.30  36.53
infant_mort (6)      317.77    18.60 -83.90
fertility    -7.30    18.60    1.52 -5.92
pop65        36.53    -83.90 (7)    50.99
> round(cor(X)[1:4, 1:4], 2)
      life_exp infant_mort fertility pop65
life_exp    1.00    -0.89    -0.83  0.72
infant_mort -0.89    1.00    0.85 -0.66
fertility    (8)      0.85    (9) (10)
pop65        (11)    -0.66    (12) 1.00
> round(apply(X, (13), sd), 2)
      life_exp infant_mort fertility pop65 urban_pct
      (14)      (15)      1.23    7.14    22.10
      gdp_pc health_exp_pct internet_pct
      28361.75      3.27      25.18
> Z <- (16)(X)
> round(cov(Z)[1:3, 1:3], 2)
      life_exp infant_mort fertility
life_exp    1.00    (17)    (18)
infant_mort -0.89    (19)    0.85
fertility    -0.83    (20)    1.00
> (21)(Z[c("Qatar", "Kuwait", "Bahrain", "United Arab Emirates"), ])
      Qatar Kuwait Bahrain
Kuwait    1.8562957
(22)      2.1483172 0.4388862
United Arab Emirates 1.7262102 0.8643204 1.1809634
> d2 <- mahalanobis(X, colMeans(X), (23))
> round(sort(d2, (24) = TRUE)[1:6], 2)
      Monaco Naoero Afghanistan Somalia, Fed. Rep.
      75.52    47.67    46.15    29.55
      Lesotho (25)
      25.60    21.15
> (26)(-d2)[["Qatar"]]
[1] 6
> XW <- X %*% solve(chol(cov(X))) # Lecture 5, the whitening line
> all.equal(as.matrix(dist(XW))["Qatar", "Kuwait"]^2,
+ (27)(X["Qatar", ], X["Kuwait", ], cov(X)))
[1] (28)
```

Answer table — Problem 3. One entry per box; write exactly what the box hides.

box	contents	box	contents
(1)		(15)	
(2)		(16)	
(3)		(17)	
(4)		(18)	
(5)		(19)	
(6)		(20)	
(7)		(21)	
(8)		(22)	
(9)		(23)	
(10)		(24)	
(11)		(25)	
(12)		(26)	
(13)		(27)	
(14)		(28)	

△! The three errors this sheet exists to prevent

1. **Reading a covariance as if it were a correlation**, or comparing variances in different units (Problem 1(d), (f)): **S** carries the units, **R** does not.
2. `mahalanobis(X, xb, solve(S))` (Problem 2(d), box (23) of Problem 3): no error, no warning, and not d^2 .
3. **A distance on raw mixed-unit columns** (Problem 2(a), (c), (d)), or on rows standardised *after* subsetting (Problem 2(c)): both run, both answer a question nobody asked. Say the dimensions and units aloud before Enter.

Quiz 1: Thursday, Sep 10, in class, Lectures 1–5, fresh data, no R.